

AE-325: Preparation of samples for ultrafast and raman spectroscopy from [Ru(CO)(OH)₂(PNPtBu)] (AE-310-2)

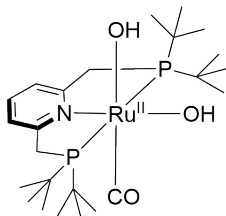
Date: 2024-07-22

Tags: PNP P(tBu)N(py)P(tBu)
[Ru(CO)(OH)₂(PNP)] AE Ultra Fast
spectroscopy Cooperation AG Dietzek-
Ivanšić

Status: Done

Created by: Alexander Eith

Reaction scheme/sample structure



Chemical Formula: C₂₄H₄₅NO₃P₂Ru
Molecular Weight: 558,6452

[Ru(CO)(OH)₂(PNP)]

Literature/reference experiments

Literature	/
Reproduction	/
Related experiment	/

Reagents

Name	CAS Number / Experiment Number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume _{theo} [ml]	Volume _{exp} [mL]
[Ru(CO)(OH) ₂ (PNPtBu)]	AE-310: Synthesis of [Ru(CO)(OH) ₂ (PNPtBu)] using Ag ₂ O, 5 h (AE-310-2)	3.75 + 4.39	/	2.094 + 2.45 + 0.4188	2.11 + 2.42 + 0.42	558.645	/	/
Milli-Q water	Experiment - GD-003: Degassing of Milli-Q-Water	/	/	/	/	/	5 + 1.4 + 4	/
KOH (1 M in water)	Experiment - AE-243: 1 M Stocksolution of KOH in water	/	/	/	/	/	0.1 + 25 μL + 0.2	/

Procedure/observations

All steps (unless stated otherwise) were carried out under argon using standard [Protocol - Schlenk Technique](#) without heating the glas ware prior to use.

Since the compound is light sensitve the steps were carried out with the light turned out in the fumehood and the lab.

Date	Time	Step	Observations
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19.07		Two 10 mL flasks were set under argon	
	10:00	To each flask [Ru(CO)(OH) ₂ (PNPtBu)] was added. To AE-325-1 2.11 mg and to AE-325-2 2.42 mg	
		The flasks were set under argon	
		The flasks were stored under Ar in the dark at -20 °C	
22.07	10:35	To AE-325-1 water (5 mL) and to AE-325-2 water (1.4 mL) was added.	before filt.jpg
		Both flasks were stirred	
		Two 10 mL Schlenks were prepared	
	10:50	Both solutions were filtered using a PA-syringe filter (0.2 µm) into new flasks each	AE-325-3 : filtrate of AE-325-1 AE-325-4 : filtrate of AE-325-2 significantly lighter in colour than before after filt.jpg
		Two 10 mL Schlenk flasks were prepared	
	10:55	To one flask AE-325-3 (1.9 mL) was added giving AE-325-5 , to the other flask AE-325-4 (0.47 mL) was added giving AE-325-6	
	11:05	To AE-325-5 KOH (0.1 mL, 1 M) was added using a 1 mL syringe and to AE-325-6 KOH (25 µL, 1 M) was added using an 50 µL hamilton syringe	Both solutions got a stonger colour and the colour became more orange
	11:15	AE-325-4 was transffered into a prepared 1 mm according to Protocol - Inert filling of Young type cuvettes provided by AG Dietzek-Ivansic	AE-325-3.jpg
	11:25	AE-325-6 was transffered into a prepared 1 mm according to Protocol - Inert filling of Young type cuvettes provided by AG Dietzek-Ivansic	AE-325-4.jpg
	11:35	AE-325-5 was transffered into a prepared 1 cm according to Protocol - Inert filling of Young type cuvettes provided by AG Dietzek-Ivansic	Some of the solution got pushed out through the Young-type valve. When closed the cuvette is still sealing and enough sample remained, therefore sample was used. AE-325-5.jpg
	11:45	AE-325-3 was transffered into a prepared 1 cm according to Protocol - Inert filling of Young type cuvettes provided by AG Dietzek-Ivansic	AE-325-6.jpg
	13:30	The samples were hand over to Mohini from AG Dietzek-Ivansic	

25.08	11:00	In a prepared flask [Ru] (0.4188 mg) was added under air	
		The flask was set under argon	
	11:45	To the flask water (1.9 mL) and subsequently KOH (1 M, 0.10 mL) was added.	AE-325-7
		The mixture was stirred shortly	
		The mixture was filtered into a new prepared flask using a PA syringe filter (pore-diameter = 0.2 µm)	
	11:51	AE-325-7 was transferred into a prepared 1 cm according to Protocol - Inert filling of Young type cuvettes provided by AG Dietzek-Ivansic	AE-325-7.jpg
	10:50	Into a prepared flask water (1.9 mL) and KOH (0.10 mL, 1 M) was added	AE-325-8
	11:35	AE-325-8 was transferred into a prepared 1 cm according to Protocol - Inert filling of Young type cuvettes provided by AG Dietzek-Ivansic	AE-325-8.jpg

Results

see

[Transient-measurements-data-2-for-Rull-jacobs-sample-MS_20240814_LZ.pptx](#) for presentation of results.

Summary by AE: no significant difference between samples in water or in KOH (50 mM).

Resonance raman at 405 nm: 1st photon absorption: MLCT to pyridine moiety, since pyridine bands are visible (first time observed) (good result)

Nanosecond emission spec. at 405 nm: sharp peak: likely water, broad peak: of interest: shows, that an emissive relaxation pathway exists (first time observed, good result)

Transient absorption spec: no good results. Issues with stability. Assigned life times cannot be trusted, also further interpretation cannot be trusted

Next steps

using rotating stage for measurements, like was done from JSN in <https://doi.org/10.1039/D1EE01053K>.

Remeasurment of [Ru(CO)(OH)₂(PNN)] in setup in Jena for better comparison

Investigation of concentration and light pathlength influence: samples in 1 mm and 1 cm cuvettes with a) same concentration, b) same optical density (make sample for 1 mm cuvette --> dilute by factor 10 make sample for 1 cm cuvette, no filter in between)

Linked experiments

- [KRA-104: Degassing of Milli-Q water](#)
- [AE-238: Synthesis of \$\[\text{Ru}\(\text{CO}\)\(\text{OH}\)_2\(\text{PNPtBu}\)\]\$ using \$\text{Ag}_2\text{O}\$, 5 h](#)
- [AE-243: 1 M Stocksolution of KOH in water](#)
- [AE-252: Synthesis of \$\[\text{Ru}\(\text{CO}\)\(\text{OH}\)_2\(\text{PNPtBu}\)\]\$ using \$\text{Ag}_2\text{O}\$, 5 h](#)
- [GD-003: Degassing of Milli-Q-Water](#)
- [AE-310: Synthesis of \$\[\text{Ru}\(\text{CO}\)\(\text{OH}\)_2\(\text{PNPtBu}\)\]\$ using \$\text{Ag}_2\text{O}\$, 5 h](#)

Linked resource

Protocol - [Inert filling of Young type cuvettes provided by AG Dietzek-Ivansic](#)

Attached files

Transient-measurements-data-2-for-Rull-jacobs-sample-MS_20240814_LZ.pptx
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AE-325-8.jpg
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AE-325-7.jpg
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AE-325-4.jpg
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AE-325-5.jpg

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AE-325-6.jpg

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AE-325-3.jpg

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after-filt.jpg

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before-filt.jpg

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Unique eLabID: 20240722-ef67908663638ed5ccb1879c0c032efbe5216905
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1223>